

Technical Market Indicators Optimization using Evolutionary Algorithm

P.Fernandez-Blanco¹, D.Bodas-Sagi¹, F.Soltero¹, J.I.Hdalgo^{1,2}

¹Ingenieria Tecnica de Informatica de Sistemas CES Felipe II de
Aranjuez, UCM

²Departamento de Arquitectura de Computadores Universidad
Complutense de madrid

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Presenter: Hung-Hsi Chen
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Abstract(1/2)

Real world stock markets predictions such as stock prices, unpredictability, and stock selection for portfolios, are challenging problems. Technical indicators are applied to interpret stock market trending and investing decision. The main difficulty of an indicator usage is deciding its appropriate parameter values, as number of days of the periods or quantity and kind of indicators. Each stock index, price or volatility series is different among the rest.

Abstract(2/2)

In this work, Evolutionary Algorithms are proposed to discover correct indicator parameters in trading. In order to check this proposal the **Moving Average Convergence Divergence(MACD)** technical indicator has been selected. Preliminary results show that this technique could work well on stock index trending. Indexes are smoother and easier to predict than stock prices. Required future works should include several indicators and additional parameters.

MACD

- **Demand Index, DI:** $DI = \frac{H + L + 2 \times C}{4}$
where H, L, C are the high, low, and closing values
- **Exponential Moving Average, EMA :**
 $EMA[t] = EMA[t-1] + \alpha(DI - EMA[t-1])$
 $\alpha = 2 / (1 + t)$ is called smoothing constant
 Ex : EMA_9 , Weight :

EMA(t-1) 20%	DI 80%
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- **Difference, DIF :** $DIF = EMA_{12} - EMA_{26}$
- **DEM, also called MACD :**
 $DEM_t = DEM_{t-1} + \alpha(DIF_t - DEM_{t-1})$
 $\alpha = 2 / (1 + t)$
- **Histogram = DIF - DEM**

TSMC daily chart



台積電

2008/12/29

開盤 43.6
最高 43.6
最低 42.6
收盤 43.35
漲跌 0.05
成交 1.76萬張

均價線

5 日 43.42-
20 日 42.29+
60 日 42.87-
120 日 49.67-

均量線

5 日 1.99萬張-
10 日 3.84萬張-

KD 指標(9 日)

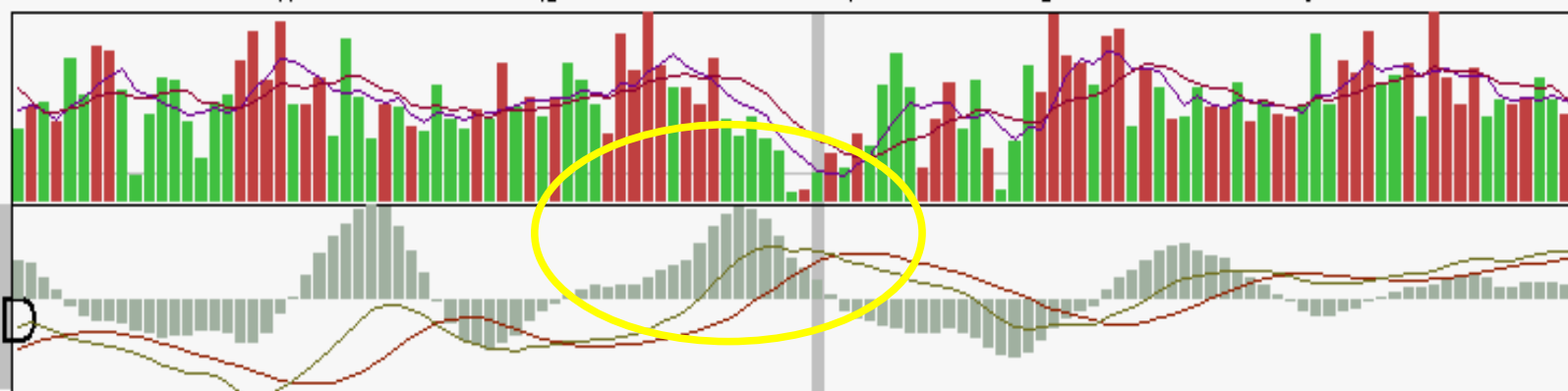
RSI 指標

MACD 指標

DIF 3-

MACD 2.35+

PSY 指標



日線圖

> 遠 近 還原 除權

TSMC weekly chart



台積電

2009/3/30

開盤 51.8
最高 52
最低 50.8
收盤 52
漲跌 0.1
成交 4.6萬張

均價線

4週 50.27+
13週 45.85+
26週 44.54-
52週 51.91-

均量線

4週 26.95萬張-
13週 28.75萬張-

KD指標(13週)

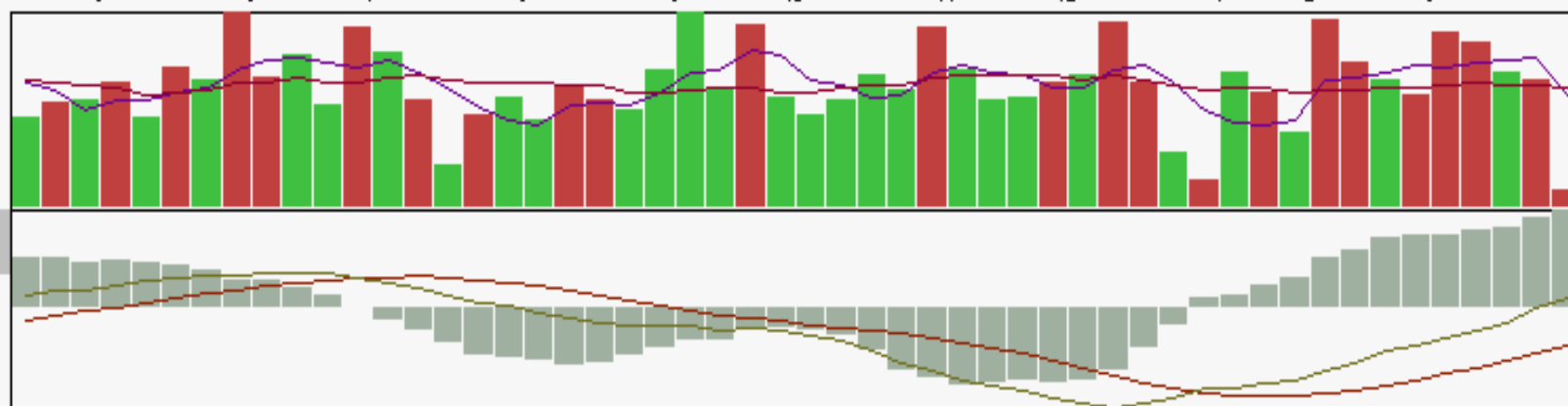
RSI指標

MACD指標

DIF 1.17+

MACD -3.91+

PSY指標



MACD

週線圖

遠 近 還原 除權

MACD(used here)

- $MACD = EMA_{12} - EMA_{26}$
- $Signal = EMA_9$
- $Histogram = MACD - Signal$

Bullish Signals :

1. Positive Divergence
2. Bullish Moving Average Crossover
3. Bullish Centerline Crossover

Bearish Signals :

1. Negative Divergence
2. Bearish Moving Average Crossover
3. Bearish Centerline Crossover

Merrill Lynch & Co., Inc. (MER) NYSE

© StockCharts.com

22-Nov-1999 4:00pm **Open** 41.68 **High** 43.02 **Low** 40.71 **Last** 42.35 **Volume** 4.2M **Chg** +0.43 ▲

1/2 Merrill Lynch 42.35 (Daily)

— EMA(12) 40.22

— EMA(26) 38.31



Positive Divergence



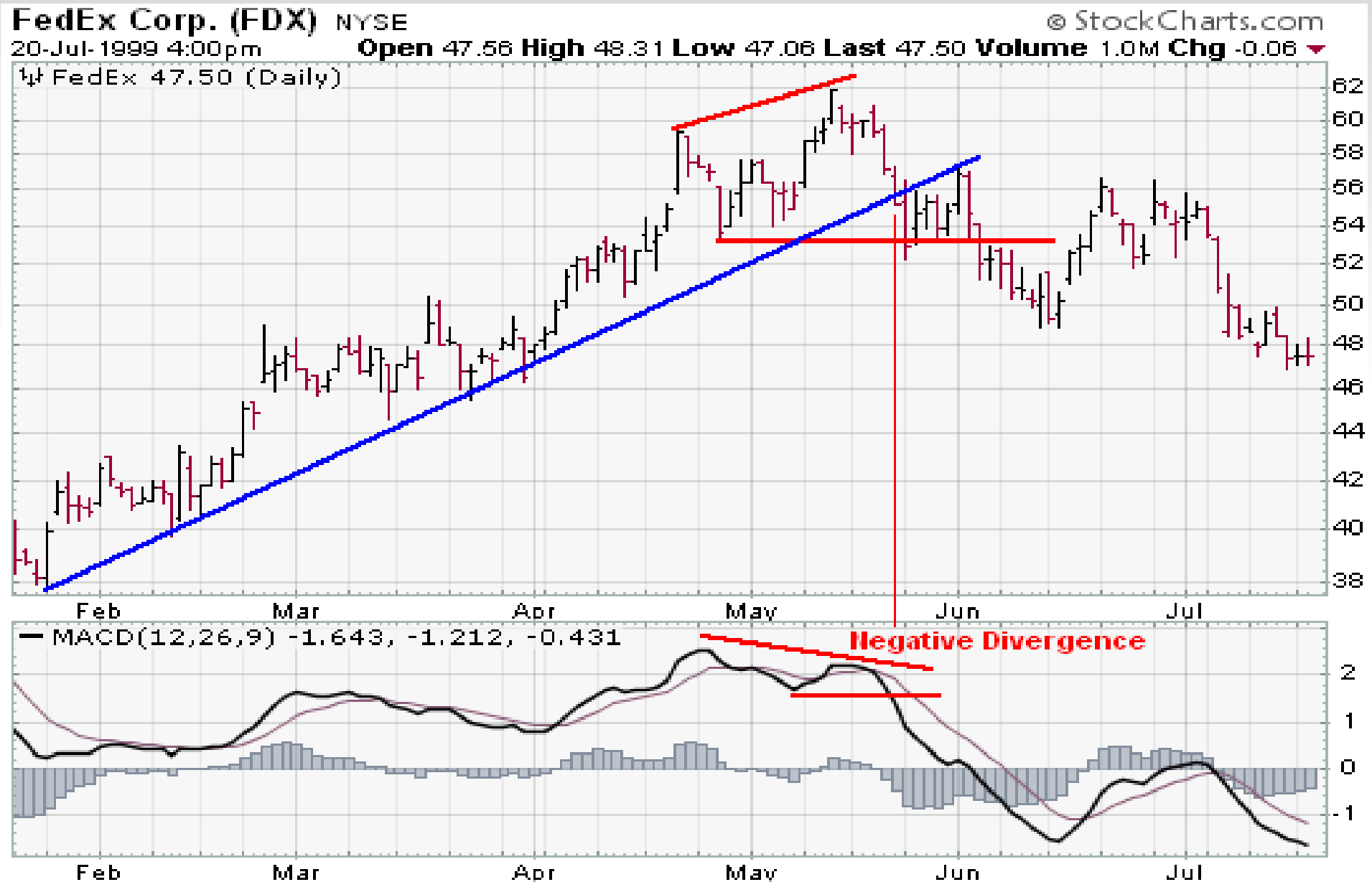
Bullish Moving Average Crossover



Bullish Centerline Crossover



Negative Divergence



Bearish Moving Average Crossover

Merck & Co., Inc. (MRK) NYSE

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15-Apr-1999 4:00pm

Open 72.88 High 73.68 Low 70.35 Last 72.28 Volume 8.8M Chg -1.05 ▼

1/4 Merck 72.28 (Daily)



Bearish Centerline Crossover

Unisys Corp. (UIS) NYSE

12-Oct-1999 4:00pm **Open** 43.12 **High** 43.69 **Low** 42.56 **Last** 43.12 **Volume** 3.6M **Chg** +0.25 ▲

Unisys 43.12 (Daily)



Experiment

- We have three parameters of the MACD
 EMA_9 , EMA_{12} , EMA_{26}
- We use Evolutionary Algorithms to find the parameters that can maximize the profit.
- **Note:** If we concern several indicators simultaneously, we can use Multilobjective Evolutionary Algorithms(MOEAs) to deal with the multiple constraints.
- Experiment on Dow Jones Industrial Average(DJIA)
(from 1/1/2000 to 12/31/2005)

Evolutionary Algorithm

- **Fitness** is calculated taking into account the maximum achievable profit.

	MACD			N
	12	26	9	20

Constraint :

- N: the amount of historical data, it must be greater than 30
- The second parameter of MACD must be the greatest
- The third parameter of MACD must be the smallest

Table 2. Evolutionary Algorithm Parameters

Number of genes	4
Population size	10
Mutation probability	0.09
Crossover probability	0.8
Local search	Every 100 generations

Dow Jones Industrial Average(1/2)

Company	Symbol	Industry
3M	MMM	Diversified industrials
Alcoa	AA	Aluminum
American Express	AXP	Consumer finance
AT&T	T	Telecommunication
Bank of America	BAC	Institutional and retail banking
Boeing	BA	Aerospace & defense
Caterpillar	CAT	Construction and mining equipment
Chevron Corporation	CVX	Oil and gas
Citigroup	C	Banking
Coca-Cola	KO	Beverages
DuPont	DD	Commodity chemicals
ExxonMobil	XOM	Integrated oil & gas
General Electric	GE	Conglomerate
General Motors	GM	Automobiles
Hewlett-Packard	HPQ	Diversified computer systems

Dow Jones Industrial Average(2/2)

Company	Symbol	Industry
Home Depot	HD	Home improvement retailers
Intel	INTC	Semiconductors
IBM	IBM	Computer services
Johnson & Johnson	JNJ	Pharmaceuticals
JPMorgan Chase	JPM	Banking
Kraft Foods	KFT	Food processing
McDonald's	MCD	Restaurants & bars
Merck	MRK	Pharmaceuticals
Microsoft	MSFT	Software
Pfizer	PFE	Pharmaceuticals
Procter & Gamble	PG	Non-durable household products
United Technologies Corporation	UTX	Aerospace, heating/cooling, elevators
Verizon Communications	VZ	Telecommunication
Walmart	WMT	Broadline retailers
Walt Disney	DIS	Broadcasting & entertainment

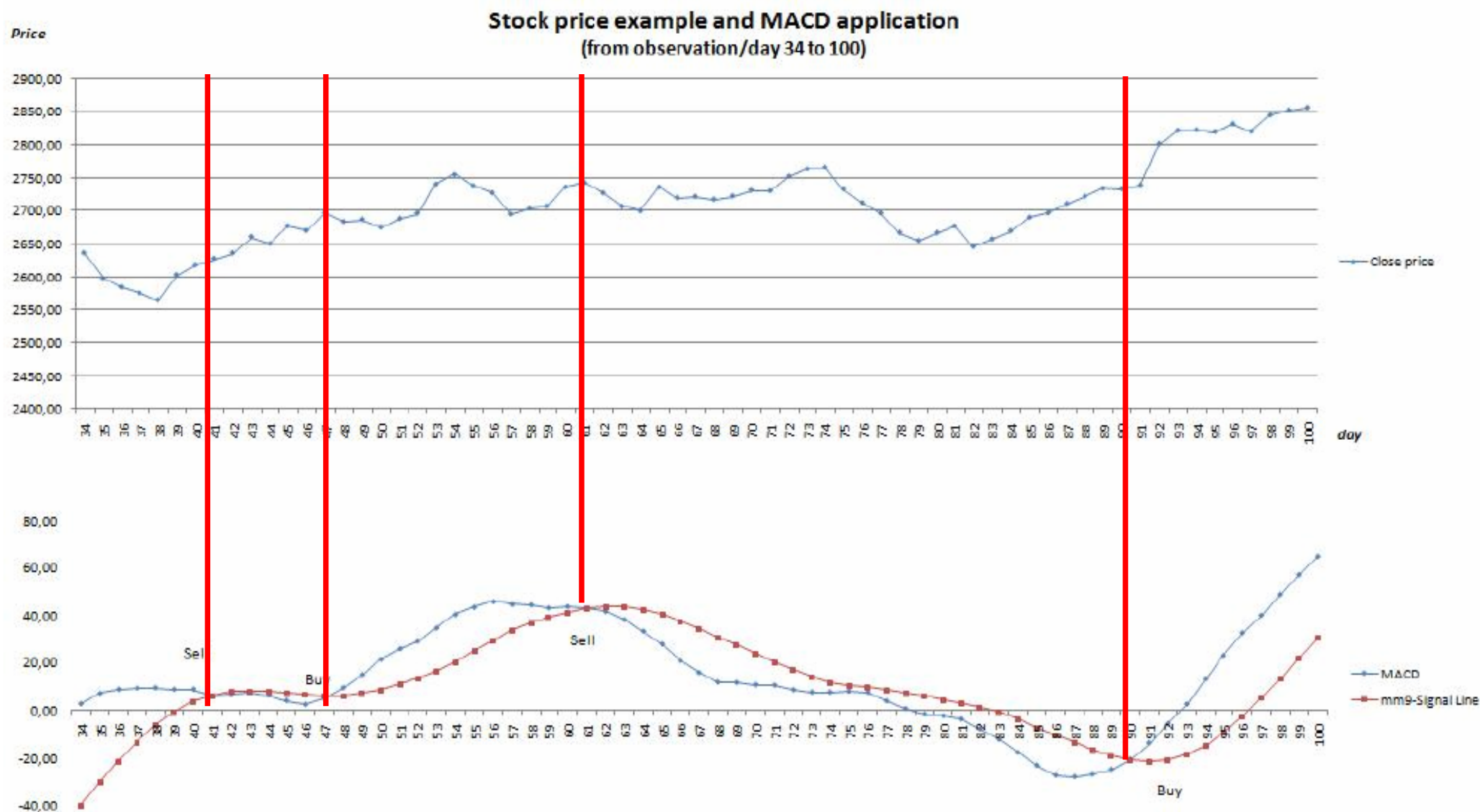


Figure 2. Stock price example and MACD application for a set of 100 days data. From observation/day 34 to 100.

Table 3. Experimental results

Number of data	Parameter 1	Parameter 2	Parameter 3	% Profit MACD	% Profit Buy and Hold
336	10	47	5	91,36	68,69
434	24	32	19	90,26	56,256
343	21	28	11	87,16	15,397
154	17	43	8	80,42	81,198
319	22	49	19	79,57	77,913
325	7	21	3	75,34	49,146
422	22	46	14	74,94	-0,23734
309	17	47	4	73,09	88,703
312	7	22	6	71,27	47,299
307	21	29	13	71,16	9,7056
228	22	33	12	70,97	80,283
424	21	35	15	70,7	49,176
382	20	38	5	70,60	42,269
231	20	39	17	68,02	75,29
439	22	50	8	67,08	47,984

Table 4. Significance of the experiments

	Mean profit	Standard deviation
Random	21,79%	14,71
EAs	50%	14,15
Buy&Hold	30%	25

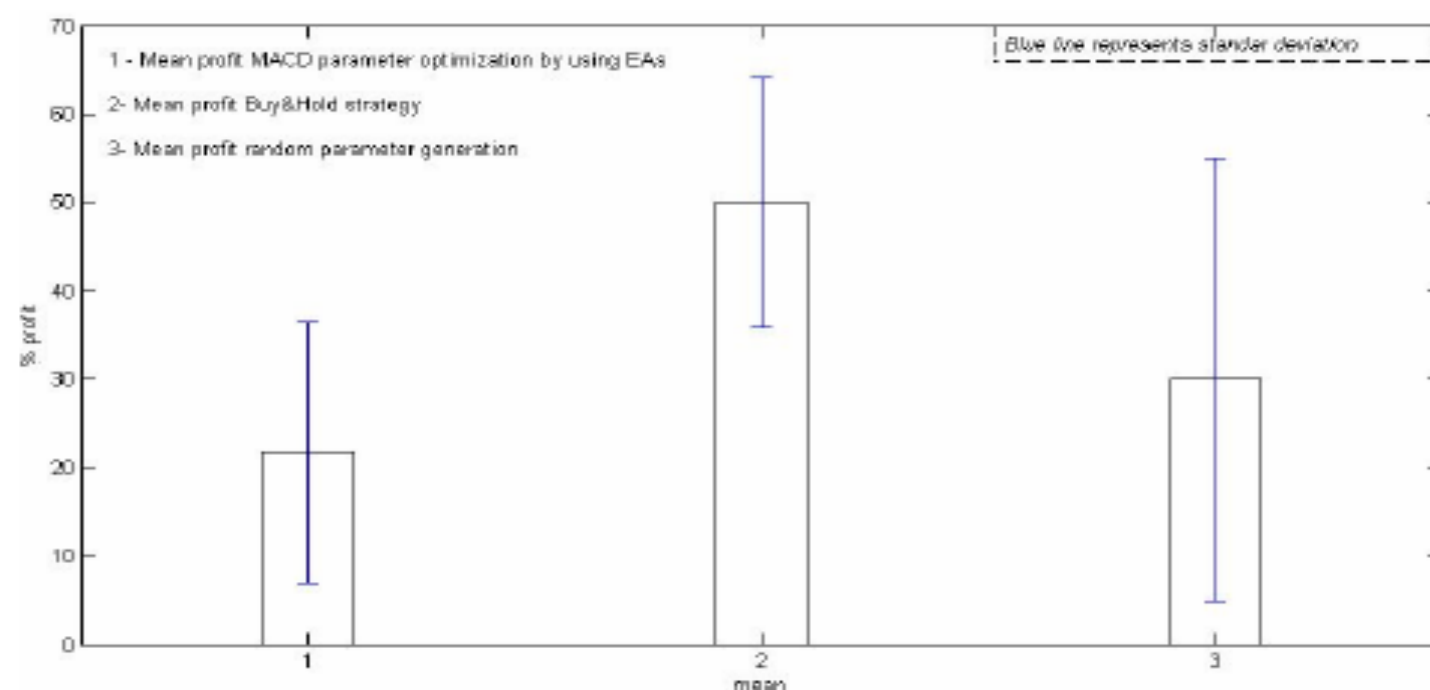
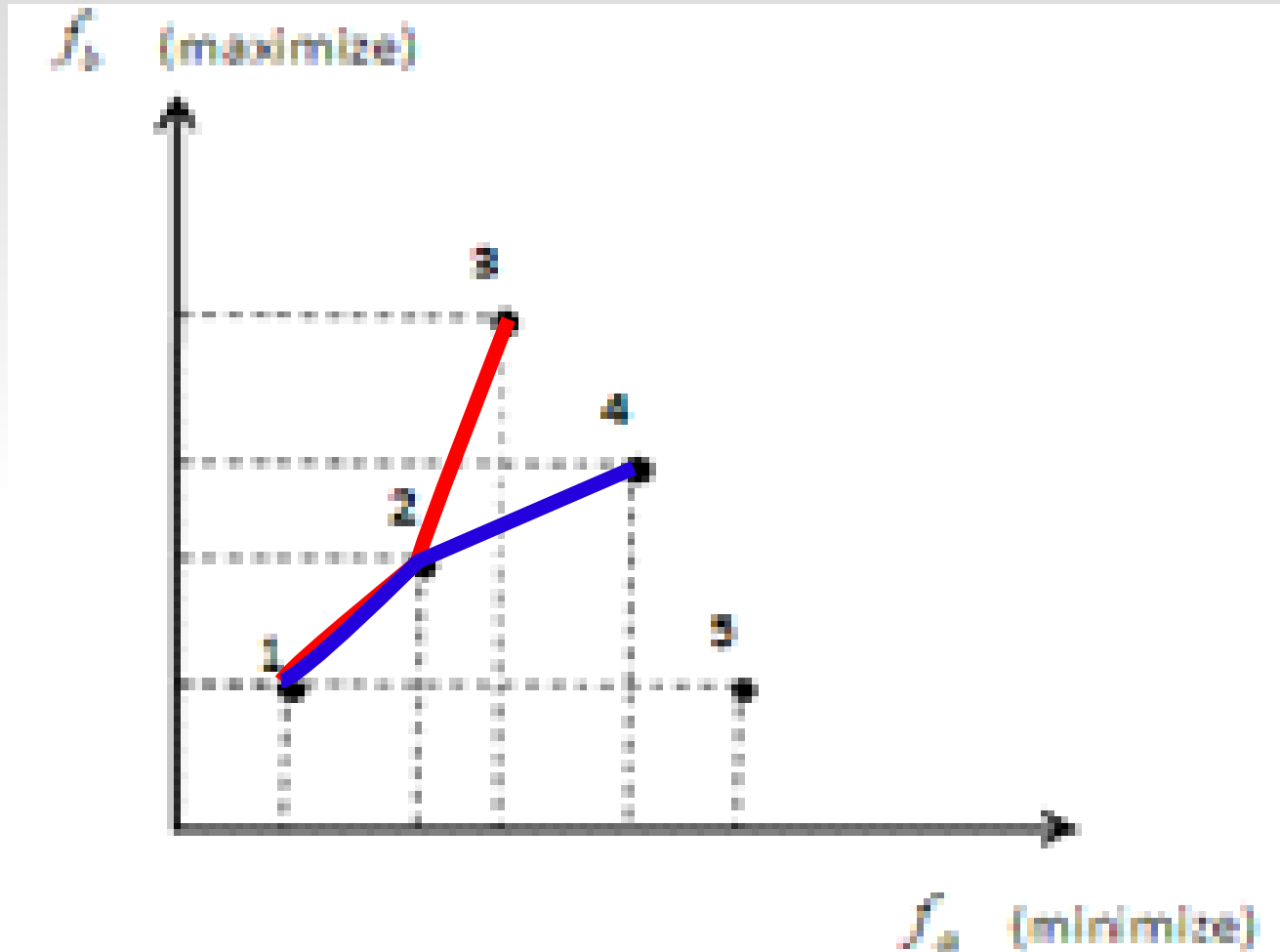


Figure 3. Means and standard deviations of MACD optimized with EAs, Buy and hold Strategy and random parameter generation.

Thank you!!

Multipleobjective Optimization

2 objectives example



The Pareto optimal front: {solution 1, 2 3}

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